# EU Diesel Rerouting

## At a glance

An end-to-end data project that quantifies how EU diesel supply was rerouted after the February 2023 embargo on Russian refined products, built entirely on public data. By Elias El Hamdaoui, junior data analyst, Geneva area.

- Code and full methodology: github.com/EliasElh/eu-diesel-rerouting
- Dashboard: 4-page Power BI report (screenshots and PDF in the repository)

## The question

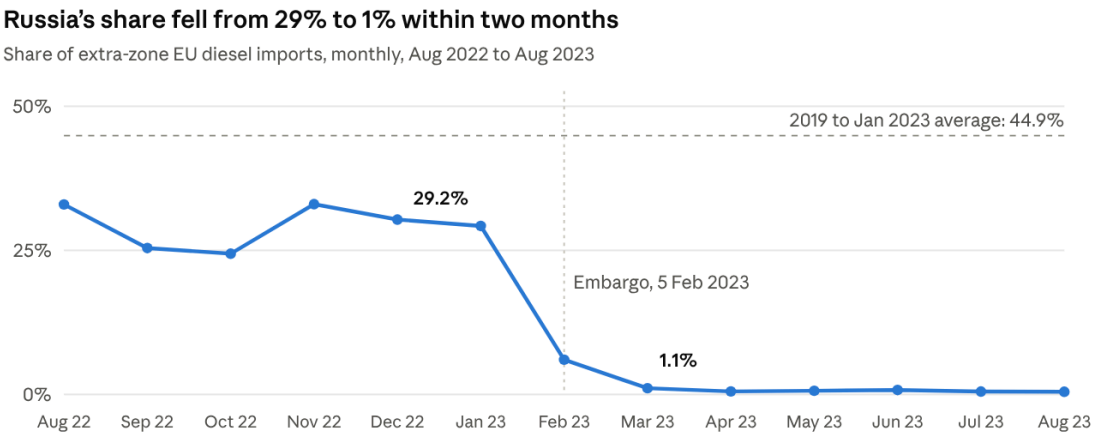
Before the embargo, Russia was the EU's largest external diesel supplier, a few days' sailing from Rotterdam. On 5 February 2023 that supply was banned. The project answers three questions:

1. **Origin shift**: who replaced Russia, and how fast?
2. **Price vs. flow**: did import volumes respond to the refining margin (crack spread)?
3. **Distance**: how much further does EU diesel now travel?

## Key findings

The switch away from Russia took one month, was driven by the embargo itself rather than prices, and made EU diesel travel 53% further.

| Question | Finding | Key figure |
|---|---|---|
| Origin shift | Russia supplied 2.5 to 4.5 Mt a month before the embargo, then collapsed within one month; Saudi Arabia, the US, Kuwait, India and the UAE replaced it simultaneously | Share of extra-zone imports: 44.9% before (2019 to Jan 2023), below 2% from Mar 2023, 0.6% average after |
| Price vs. flow | No significant correlation between the crack spread and import volumes at any lag; the crack spread peaked in Oct 2022, months before the embargo, so the shift reads as a regulatory shock, not a price response | Max $\|r\|$ = 0.17 on levels, 0.18 on monthly changes, lags of 0 to 6 months |
| Distance | Replacing nearby Russian supply with Gulf, Indian and US cargoes made each tonne travel much further | Average distance 4,509 to 6,905 nautical miles (+53%); tonne-miles 34.9 to 45.2 billion a month (+29.5%) |




Project pipeline output (docs/market_shares.csv) · Russia's monthly share of extra-zone imports, Aug 2022 to Aug 2023

## Method and stack

A reproducible pipeline from public APIs to a Power BI report, rerunnable end to end from the repository.

- **Data**: UN Comtrade monthly imports of HS 271019 (diesel proxy) declared by 30 countries (EU-27, UK, Norway, Switzerland), Jan 2019 to Dec 2025, 240 API calls; US EIA Brent and NY Harbor diesel prices; sea distances for the 18 origins covering 95.6% of extra-zone volume
- **Pipeline**: Python (requests, pandas) with idempotent ingestion and raw/clean separation, Parquet, DuckDB for SQL analysis (CTEs, window functions, views)
- **Reporting**: Power BI star schema around a date dimension, DAX measures (CALCULATE, DIVIDE), 4 report pages

## Data quality work

Each issue below was found by a consistency check built into the pipeline, then fixed and documented.

- **Double counting**: totals and their transport, customs-regime and consignment breakdowns were published side by side; only total rows are kept
- **Inconsistent totals**: a 14% gap against reported "World" totals was traced to zero-weight World lines from France, the UK and Spain; after the fix, the gap is 0.000% on 2,374 comparable reporter-months
- **Missing zeros**: Comtrade omits zero flows, so zero rows were rebuilt, but only for months a country actually reported
- **Transit hubs**: 60% of Dutch and 77% of Belgian declared imports come from other European reporters (Rotterdam and Antwerp transit), so origins are measured outside the 30-country zone

## Limitations

The figures are robust in direction and order of magnitude, with these documented boundaries:

- HS 271019 is a proxy for diesel: it also includes some heating oil and jet fuel
- Import declarations only: Russia stopped publishing detailed export data after 2022
- Austria excluded (no monthly data for most of the period); 2026 excluded (incomplete reporting)
- Some countries declare value without weight; the weight is then estimated by the UN (flag kept in the data)
- Extra-zone totals may slightly double-count cargoes re-declared after transit inside Europe
- Diesel price is a US benchmark (NY Harbor) used as a global proxy, not a European quote
- Distances use one port per country, routed to Rotterdam: 5 verified (67% of volume), 1 substitute port (India, 9%), 12 estimated (20%); the remaining 4.4% of volume is excluded